

Psychological Tests: Why Do We Need Them?

“Psychology is the scientific study of behavior and mental processes Human or Animal” (Feldman) There are three important terms used in this definition.....scientific study, behavior, and mental processes.

Behavior and mental processes constitute the content or subject matter of psychology, whereas scientific study refers to the methodology used by psychologists. Scientific method ensures that the results or conclusions of an investigation are objective and consistent.

For this purpose psychologists use carefully designed tools of data collection. Psychological tests are one of those tools,

We had discussed that the main and important goals of psychology, or in other words of understanding human behavior and mental processes, are;

- a) To understand the nature and mechanisms of behavior and mental processes
 - b) To develop an understanding of the relationship between behavior and mental processes
 - c) To apply this understanding to real life situations and, on the basis of this understanding, predict for the future
 - d) To employ the scientific approach for developing this understanding
- We also studied that the main goals of psychology can be categorized as:
- Observation,
 - Description
 - Understanding
 - Explanation
 - Prediction
 - Control of human behavior and mental processes
- Psychological tests help and assist psychologists in attaining all of these goals,

What is a psychological assessment?

Psychological assessment is a series of tests conducted by a psychologist, to gather information about how people think, feel, behave and react. The findings are used to develop a report of the person's abilities and behavior—known as a psychological report—which is then used as a basis to make recommendations for the individual's treatment.

Psychological assessments and reports are used in other fields as well—like in the case of career planning for young adults or in the job application process to determine how well an applicant will fit into the position.

The procedures used to create an assessment are:

- Interviews
- Observation
- Written assessment
- Consultation with other mental health professionals
- Formal psychological tests

What is a psychological test?

A psychological test is used to measure an individual's different abilities, such as their aptitude in a particular field, cognitive functions like memory and intelligence or even traits e.g. introvert and extrovert etc. These tests are based on scientifically tested psychological theories.

The format of a test can vary from pencil and paper tasks to computer-based ones. They include activities such as puzzle-solving, drawing, logic problem solving, and memory games.

Some tests also use techniques—known as projective techniques—which aim to access the unconscious. In these instances, the subject's responses are analyzed through psychological interpretation and more complex procedures than the non-projective techniques mentioned above. For example, the Rorschach test, popularly known as the ink-blot test can provide insight into the person's personality and emotional functioning.

Psychological tests may also involve observing someone's interactions and behavior. Based on the result of the test, conclusion will be drawn about the individual's abilities and potential.

Cohen and Swerdlik (1999) have enlisted a number of assumptions that are basic to psychological testing and assessment:

Psychological traits and states exist:

Traits refer to characteristics or psychological features that differentiate one person from another. According to Guilford (1959, p. 6) a trait is “any distinguishable relatively enduring way in which one individual varies from another”.

Psychological traits and states can be quantified and measured

Quantities of course refer to numbers and quantification means that when we use psychological test we gather and present the findings in terms of numbers.

Quantification makes the whole process objective. Also, quantities are comparable. If quantitative data regarding a particular trait are available then one can compare the scores of various people and assess as to who is higher and who is lower; who is more intelligent and who is less intelligent; who has depressive tendencies and who does not. The second concept in this assumption is that of measurement; it can be defined as “the act of assigning numbers or symbols to characteristics of subjects (people, events, whatever) according to rules” (Cohen & Swerdlik, 1999, p. 19). For the sake of quantitative measurement, we use scales. A scale, according to the same author, is “a set of numbers (or other symbols) whose properties model empirical properties of the objects or traits to which numbers are assigned

Difference between test and assessment

- A test is a measurement device or technique used to quantify behavior or aid in the understanding and prediction of behavior”
- Assessment includes more than just tests. A typical assessment plan may include a test or a battery of tests, interview, behavioral observation, and case history data

Uses of Psychological Test

1. **Understanding People:** They help understand how people think, feel, or behave.
 2. **Comparing Differences:** These tests compare how individuals are different from each other.
 3. **Measuring Changes:** They measure how someone might change over time in their thoughts or feelings.
 4. **Assessing Abilities:** They check what someone is good at or what they might struggle with, like in school or work.
 5. **Diagnosing Problems:** Psychological tests help identify issues like learning disabilities, emotional struggles, or mental health conditions.
 6. **Guiding Decisions:** They assist professionals, like psychologists or counselors, in making decisions about treatment or support for someone.
 7. **Assessment of Skills and Abilities:** Evaluating strengths and weaknesses in various areas, such as learning, memory, or problem-solving.
 8. **Identifying Emotional Health:** Assessing emotions and mental health conditions like anxiety, depression, or stress.
 9. **Measuring Personality Traits:** Analyzing different personality aspects such as introversion/extroversion, conscientiousness, or openness to experience.
 10. **Diagnosing Disorders and Conditions:** Detecting conditions like ADHD, autism, or specific learning disabilities.
 11. **Research and Development:** Contributing to research by providing data and insights into human behavior, cognition, and emotions for developing new theories or treatments.
- ### Types of Psychological tests
1. Individual versus group tests
 2. Ability, achievement, or aptitude tests
 3. Intelligence versus personality tests

4. Speed test versus ability tests
5. Structured personality tests versus projective tests
6. Verbal versus non-verbal/performance tests.
7. Commercial copyrighted tests versus available to all test **Individual versus Group**

Tests:

Some tests are for one person at a time (like WAIS, WISC), while others can be done with many people together (like Raven's Progressive Matrices). Individual tests need one examiner for one person, while group tests are quicker, with multiple people and often use multiple-choice questions. Group tests are easier to score and can be administered through recorded instructions or computers. However, they might not be ideal when a good connection between the person being tested and the examiner is necessary.

Ability, achievement, or aptitude tests

Ability tests, like intelligence tests, measure your thinking and learning skills, providing an IQ score. They show what you know at a certain time and your level of skills in different areas.

Achievement tests check what you've learned from specific training or programs, like SATs at the end of courses, to see if you met the goals.

Aptitude tests measure the overall effect of your life experiences on your learning abilities. They predict how well you might do in the future, unrelated to specific training. Differential Aptitude Tests (DAT) are common examples.

Intelligence Verses Personality Tests:

Even though the test names suggest what they measure, it's important to know that personality tests don't tell you about IQ, and IQ tests can't tell you about personality traits. Many confuse these, but they're different and come in many varieties, both being widely available.

Speed Tests versus Ability Tests:

In ability tests, like IQ tests, the focus is on measuring the level of ability, not how fast you finish. Some tests, such as Raven's Progressive Matrices, don't have a time limit. However, in other tests like Clerical Speed and Accuracy Test, how quickly you perform is important. Power tests have a time limit, but it's long enough for everyone to finish.

Structured Personality Tests versus Projective Tests:

Personality tests come in different types and measure usual behavior like traits. There are two main categories: structured tests and projective tests. Structured tests have fixed answer options, like true/false or multiple-choice questions, where you pick the one that best describes you. These tests are often self-report type.

Projective tests, unlike structured ones, are harder to score and interpret. They involve showing vague images or words to people who then describe what they see or complete the task. For example, Rorschach's Inkblots or TAT, where people tell stories about pictures, are used to understand personality. Other tests, like HTP or RISB, involve drawing or completing sentences to reveal traits or tendencies.

Verbal versus Nonverbal/ Performance Tests:

Many tests rely on language skills, especially IQ tests and structured personality tests. However, tests like Raven's Progressive Matrices don't measure verbal abilities or use language.

Commercial-Copyrighted Tests versus "Available To All"/Online Tests:

Most copyrighted IQ tests, like WAIS or WISC, can't be copied or used without permission as it's unethical and diminishes their value if people already know the content. However, some tests available online or in books are mainly for research, not for diagnosing or screening. Personality tests are more commonly found online, like the Multidimensional Health Locus of Control Scale or Self Efficacy Scale, which authors have made available for use, downloading, and printing, offering valuable research evidence.

Ethical, Legal and Professional issues in assessment

1. **Confidentiality and Privacy:** Safeguarding sensitive information collected during assessments to ensure it remains private and is shared only with authorized individuals or for authorized purposes.
2. **Informed Consent:** Obtaining explicit consent from participants or individuals being assessed, providing comprehensive information about the assessment process, its objectives, and how the results will be used.
3. **Fairness and Equity:** Ensuring assessments are conducted impartially, free from bias or discrimination based on factors like race, gender, ethnicity, or socioeconomic status.
4. **Validity and Reliability:** Ensuring assessment tools accurately measure what they are intended to measure and produce consistent, dependable results.
5. **Professional Competence:** Conducting assessments within one's area of expertise, adhering to professional standards, and maintaining ongoing professional development.

6. **Legal Compliance:** Abiding by laws and regulations governing assessments, such as those related to data protection, disability accommodations, and informed consent.
7. **Avoiding Conflicts of Interest:** Preventing situations where personal interests or biases could influence assessment outcomes, ensuring objectivity and fairness.
8. **Appropriate Use of Results:** Using assessment outcomes responsibly, ensuring accurate interpretation, and avoiding misuse that could have adverse consequences for individuals.
9. **Cultural Sensitivity:** Recognizing and addressing cultural influences that might affect the assessment process or interpretation of results, ensuring assessments are culturally relevant and unbiased.
10. **Clear Communication and Feedback:** Providing understandable and meaningful feedback to individuals being assessed, ensuring they comprehend the results and their implications for further action or support.

Adherence to these ethical, legal, and professional principles is vital in maintaining integrity, fairness, and respect for individuals throughout the assessment process.

Characteristics of a Test

□ Reliability

What Is Reliability in Psychology?

According to Kaplan & Saccuzzo (2001) reliability is “the extent to which a score or measure is free from measurement error. Theoretically, reliability is the ratio of true score variance to observed score variance.”

Measurement Error

The classical test score theory implies that everyone can obtain a true score on any test or measure if the measure is free of error. But we know that there is perhaps no measure that can be considered as totally error free. There is always some chance of error. This further implies that the scores that we obtain on different measures are not the true scores of the test taker. These are the observed scores plus error. In other words the observed score is not the true score of a test taker, implying that the observed score may not be a 100% true representative of a person's traits or abilities. Here we need to realize that the term 'error' is not used to indicate that something has gone wrong or we have made some 'mistake'. Error, in this context, refers to the amount and extent of variance that may be expected in results. The observed score is therefore the sum of true score and error i.e.

$$X = T + E$$

Here X refers to the observed score, T is the true score, and E is the error that can be expected in the measurement

Sources of Error in Test Scores: Psychologists and educationists try to estimate the amount and degree of error that may be expected in their measurement. That is the main reason why test developers and test administrators emphasize on uniform testing procedures besides controlling other possible sources of error.

The following are a few of the variables that may be possibly causing error in measurement:

A. Test Related Factors:

- Difficulty level (too difficult or too easy)
 - Length of the test (too long causing fatigue or boredom)
 - Domain or content (not suitable for all test takers.....suitable for some but not all)
 - Items may not be representing the domain or content
 - Time limit (may cause stress or a handicap)
- B. Test Administration Process:
- Poor and not uniform testing conditions (physical setting and environment)
 - Poor, faulty, improperly worded, improperly delivered, not uniform instructions
 - Test administrator's personality (different administrators in different situations having different personality styles)
 - Rapport (poor, too much, or too little)
- C. Examinee Related Variables:
- Prior learning and experience
 - Individual differences (within group differences; Personality styles, stress tolerance level, IQ level, emotionality, motivation, knowledge)
 - Difference from the normative sample (no group is identical to the normative sample; every group is different from every other group)
 - Within- person differences (the same persons may change over time (life, academic, and professional experiences; physical conditions, health, motivational level, emotional state etc.)

Correlational and reliability

Correlation shows the relation between 2 variables, and its measure by a statistics called the correlation of coefficient.

- Coefficient of correlation tells us two things about a relationship; the magnitude of relationship and the direction of relationship. Magnitude means the size of correlation whereas direction indicates whether the correlation is positive or negative.

- The size of correlation range between -1 (perfect negative) and +1(perfect positive) with a zero (no correlation at all) value in between
- The value of coefficient of a correlation is always 1 or less than 1
- The closer is a value of correlation to one, the stronger is the correlation e.g. $r = 0.9$.
- The closer a coefficient of correlation is to zero, the weaker will be the relationship.

Types of Reliability

According to the definition given by Anastasi & Urbina (2007) reliability refers to “the consistency of the scores obtained by the same persons when they are reexamined with the same test on different occasions or with different sets of equivalent items, or under other variable examining conditions”. There are two main types of reliability: internal reliability and external reliability.

1. **Internal reliability** means that the measure has consistency within itself. In other words, the same question posed differently would produce the same results. It is often measured using the split-half method.
2. **External reliability** refers to how results compare to results between individuals and across time. It is often measured using test-retest, inter-rater and parallel forms methods.

Measurement of Reliability

Test-Retest Reliability

Test-retest reliability is a measure of the consistency of a psychological test or assessment. This kind of reliability is used to determine the consistency of a test across time. Test-retest reliability is best used for things that are stable over time, such as intelligence.

Test-retest reliability is measured by administering a test twice at two different points in time. This type of reliability assumes that there will be no change in the quality or construct being measured. In most cases, reliability will be higher when little time has passed between tests.

The test-retest method is just one of the ways that can be used to determine the reliability of a measurement. Other techniques that can be used include inter-rater reliability, internal consistency, and parallel-forms reliability.

A major **advantage** of this method is that there is maximum control over the test taker and test item variables. The subjects are the same and the test items are also the same.

Drawbacks

1. **Changing Conditions:** Sometimes, the environment or circumstances might change between two tests. Things like noise or temperature can affect how well someone performs.
2. **Changes in People:** Test takers might change too. Their feelings, motivation, social situations, or even their health might be different the second time they take the test.

3. **Practice Effect:** Doing the same test twice might make it less challenging or more familiar for the person. Or it might become boring. This can affect their performance.
4. **Learning from the Test:** Some tests are such that people might remember the questions or figure out how to answer them better the second time. This can especially affect tests that measure thinking skills like problem-solving, reasoning, or math.
5. **Time Gap Matters:** The time between the two tests is important. If the gap is short, the results might be more similar. But if it's longer, the results might differ more. Test makers should mention the time gap when they talk about how reliable the test

Test- retest reliability is denoted as **rtt**.

Inter-Rater Reliability

This type of reliability is assessed by having two or more independent judges score the test. The scores are then compared to determine the consistency of the examiners estimates.

One way to test inter-rater reliability is to have each examiner assign each test item a score. For example, each examiner might score items on a scale from 1 to 10. Next, you would calculate the correlation between the two ratings to determine the level of inter-rater reliability.

Another means of testing inter-rater reliability is to have examiners determine which category each observation falls into and then calculate the percentage of agreement between the examiners. So, if the examiners agree 8 out of 10 times, the test has an 80% inter-rater reliability rate.

Parallel-Forms Reliability / Alternate-Form

Parallel-forms reliability is assessed by comparing two different tests that were created using the same content. This is accomplished by creating a large pool of test items that measure the same quality and then randomly dividing the items into two separate tests. The two tests should then be administered to the same subjects at the same time.

Independently developed and completely parallel or equivalent forms of the same measure. They should match each other in all respects including:

- Same specifications
- Same instructions
- Same time limit
- Same content
- Same number of items
- Same item format
- Difficulty level

Consistently and stability across the parallel of forms

Consistency over time

	Form A	Form B
Occasion 1	Group I (50% of total)	Group 2 (50% of total)
	Interval	
Occasion 2	Group 2 (50% of total)	Group I (50% of total)

Advantages

1. **Increases Confidence:** Using two different tests boosts confidence in the accuracy of results.
2. **Minimizes Memory Effects:** Prevents people from recalling answers between tests, ensuring more reliable results.
3. **Provides Diverse Insights:** Different tests offer a broader understanding of someone's abilities.
4. **Identifies Test Issues:** Consistent results indicate a reliable test, while differences might signal problems.
5. **Promotes Fairness:** Using different forms ensures fairness without favoritism for specific questions or formats.
6. **Enhances Flexibility:** Having various test versions can be advantageous in different situations.

Drawbacks

- **Hard to Create:** Making two tests that are equally good at measuring the same thing is tough. They need to be equally challenging and cover the same topics.
- **Difficult to Manage:** Giving two different tests to the same group can be tough to organize, especially in big groups or when testing conditions need to be the same.
- **Small Differences:** Even if we try hard, there might be tiny differences between the two tests that affect how well they measure the same thing.
- **Not Always Possible:** Some things are hard to test with two different versions. For example, certain skills or abstract ideas might not easily work with different test forms.
- **Practice Effect:** If people get used to the type of questions on the first test, they might do better on the second test just because they're more familiar with it. This can mess up the reliability of the test.

- Takes Time and Money: Creating and checking two versions of a test needs a lot of time, effort, and money. This might not be possible or practical in some situations.

Split half Method Reliability

This form of reliability is used to judge the consistency of results across items on the same test.

Essentially, you are comparing test items that measure the same construct to determine the tests internal consistency. It is often referred to as the split-half method of measuring reliability.

When you see a question that seems very similar to another test question, it may indicate that the two questions are being used to assess reliability.

Because the two questions are similar and designed to measure the same thing, the test taker should answer both questions the same, which would indicate that the test has internal consistency.

The test can be divided in many ways:

- Divide test into equal number of items.

The first 50% items are considered as one half and the remaining 50% as the other half. Scores are calculated separately for the two halves. There is one limitation of dividing the test this way. There is likelihood that the two halves may be having different difficulty levels. The initial items may be easier than the items in the second half. Also, items in the two halves formed in this way may pertain to different content areas

- A better option can be to consider odd numbered items in one half, and the even numbered items in the other half.

This formula makes it possible to estimate the internal consistency reliability of the test from the reliability of the two halves of the test. The formula is used for the computation of reliability while taking care of the fact that a test has been shortened or lengthened. The formula also considers the number of times the test has been shortened or lengthened.

$$r_{nn} = \frac{nr_{tt}}{1 + (n - 1)r_{tt}}$$

Spearman- Brown formula is given below:

r_{nn} = estimated coefficient r_{tt} = obtained coefficient n = number of times the test has been

lengthened or shortened e.g. two times, three times etc

In split-half reliability the test is reduced to half, therefore the reliability of the full test is calculated using the Spearman-Brown formula that involves doubling the length of the test, as follows:

$$r_{tt} = \frac{2r_{hh}}{1 + r_{hh}}$$

Draw backs

1. **Splitting Impact: Reliability** varies based on how the test is split.
2. **No Item-Level Assessment:** Doesn't gauge consistency within individual questions.
3. **Small Group Challenges:** Less accurate with fewer test-takers.
4. **Accuracy Risks:** Short tests may give unreliable reliability estimates.
5. **Assumption of Equal Difficulty:** Assumes both halves are equally challenging.
6. **Chance Influence:** Splits may sometimes result in random outcomes.
7. **Limited for Diverse Tests:** Not ideal for tests with varied question types.
8. **Length Influence:** Longer tests tend to offer more reliable estimates.
9. **Mixed Subject Disadvantage:** Might not accurately reflect overall consistency in tests covering different subjects.

Kuder- Richardson Reliability:

The Kuder-Richardson method is somewhat like the split-half method because both check how reliable a test is using just one administration. But, they're different in how they look at consistency. Instead of splitting the test into halves, the Kuder-Richardson method looks at how each question performs. Basically, it's the average of all the split-half reliability scores you can get from a test.

$$r_{tt} = \left(\frac{n}{n-1} \right) \frac{SD_t^2 - \sum pq}{SD_t^2}$$

r_{tt} = Coefficient of reliability of whole test

n = the number of items in the test

SD_t^2 = Standard deviation of total scores on the test is SD_t , squared (=variance)

p = proportion of persons who pass each item

q = proportion of persons who do not pass each item

pq = product of (p) and (q) for each item

Σpq = The sum of products of (p) and (q) for all item

- Only used for that items that their answers are either right or wrong / zero or one
- Cannot use with Likert scale and having weightage

Coefficient Alpha:

Coefficient alpha is another method, much like Kuder-Richardson, used to check how consistent a test is. But while Kuder-Richardson works only for tests where answers are either right or wrong (like true/false), coefficient alpha works for tests where answers get different scores, like in personality tests where answers have different weights. This formula helps measure the consistency of these types of tests.

$$r_{tt} = \left(\frac{n}{n-1} \right) \frac{SD_t^2 - \sum (SD_i^2)}{SD_t^2}$$

r_{tt} = Coefficient of reliability of whole test

SD_t^2 = Standard deviation of total scores on the test is SD_t , squared (=variance)

SD_i^2 = Variances for item scores

$\sum (SD_i^2)$ = sum of the variances of item scores

Standard error of measurement

The standard error of measurement (SEm) estimates how repeated measures of a person on the same instrument tend to be distributed around his or her "true" score. The true score is always an unknown because no measure can be constructed that provides a perfect reflection of the true score.

The Standard Error of Measurement (SEM) is a measure of how much mistakes or variability exists in a test. It's like a measure of the "ups and downs" in test scores for the same person taking the test multiple times.

To figure out SEM, you use a formula. If you know the reliability of the test (how consistent it is), and the standard deviation of the test scores, you can calculate SEM using a simple formula.

The formula uses the test's reliability and the variation in scores to estimate the amount of error or variability in the test results.

$$SEM = SD_t \sqrt{1 - r_{tt}}$$

SDt = Standard deviation of the test score

Rtt = reliability coefficient

For example, if a test had a standard deviation of 12 and the reliability coefficient is .90, then SEM will be $12\sqrt{1 - .9} = 3.79$. The SEM is the standard deviation of the hypothetical distribution. If XYZ attains an average IQ of 90 in the hypothetical situation it will be her true score that may range between $90 - 3.79$ and $90 + 3.79$. We remember that in normal distribution about 68% of people fall between -1 and +1 standard deviations from the mean. Therefore applying the same concept we can say that 68% of people may be expected to score within this range of scores.

Validity

Traditionally, the validity of a test has been defined as the extent to which a test measures what it was designed to measure.

What a test measures and how well it measures

The extent to which a test measures the quality it purports to measure. Types of validity evidence include content validity, criterion validity, and construct validity evidence

Psychological tests are developed primarily for two reasons:

The validity of a test should not be taken casually, and cannot be reported in general terms either. Validity is not, and should not be measured and reported as “highly” valid or “low” validity. It is reported in specific terms.

- Relationships between the test content (items) and the domain (area) that it represents
- Relationship between the test content and the objectives (purpose) that lead to test construction
- Relationship between performance (r/s) and performance on other independent measures that are known to measure the same phenomenon/trait/ability/domain etc

Psychological tests are developed primarily for two reasons:

- Achievement testing: In order to see if a specific content area has been learnt and/ or mastery is acquired, and

- Predictive function: predicting future performance on the basis of the present test's performance.

Measurement of validity

There are four type of validity

1. Construct validity
2. Content validity
3. Face validity
4. Criterion validity

- **Content Validity Evidence:**

This process is about figuring out what a test really measures by doing different studies. The researcher defines what they want to measure and creates a test for it. By looking at how well the test matches up with other things, like other tests or measurements, they can understand what the test actually means.

- **Construct Validity**

A process used to establish the meaning of a test through a series of studies. To evaluate evidence for construct validity, a researcher simultaneously defines some construct and develops the instrumentation to measure it. In the studies, observed correlations between the test and other measures provide evidence for the meaning of the test

- **Criterion validity**

“The evidence that a test score corresponds to an accurate measure of interest. The measure of interest is called the criterion

Content Validity

- A systematic examination of the test content is made in order to determine whether it covers a representative sample of the behavior domain to be measured.
- Such validation procedures measure how well the individual has mastered a specific skill or course of study

Content validity is built into a test from the outset through the choice of appropriate items. For educational tests, the preparation of is preceded by a thorough and systematic examination of relevant course syllabi and textbooks, as well as consultation with subject matter experts. On the basis of the information thus gathered, test specifications are drawn up for the item writers

Certain considerations are needed for content validation measure:

- The test should cover all major aspects and items in correct proportion (the overloading and underrepresentation of elements should be avoided).
- The domain under consideration must be fully described in advance rather than after test preparation.
- Content should be broadly covering the major objectives, application and interpretation and also factual knowledge

Specific Procedures:

The choice of appropriate items is important in content validity. A number of careful decisions are taken at the time of test development e.g.

Appropriate selection, appropriate formation, appropriate wording, and appropriate order of the items.

For educational tests, items are prepared from relevant course syllabi and textbooks and in some cases consultation with subject experts is used

Test specifications are given to item writers

- It includes instructional objectives, relative importance of topics/processes.
- It should clearly indicate the number of items for each topic.
- It may also provide sample material.
- The process of content validation should include description of all procedures in the manual; that ensure that test is appropriate and representative.

For example in subject-specialists has participated, the names, qualification and number of individuals should be stated.

The test total score and achievement score of an individual can be checked for grade progress

Application

- Content validity is basic to the validity of educational and occupational achievement tests.
- This validity measure is applicable to occupational tests designed for employee selection and classification.
- For aptitude and personality test content validation is inappropriate.

Content Validity versus Face Validity

We should not confuse content validity with face validity. Many students of psychology take them to be one and the same, whereas actually they are different.

Face validity is about the test appearing to be valid. It is not validity in the technical sense; nor is it technically measured.

Face validity is defined as “The extent to which items on a test appear to be meaningful and relevant, actually not evidence for validity because face validity is not a basis for inference” (Aiken, 1994, p.636). Face validity may be used where prediction r inference are not involved e.g. questionnaire or scale for a survey.

Criterion Validity

Whenever we plan and design a test we have a certain standard in mind that we want to meet.

- To look for a correlation between the test and another related test.
- We compare the results of our test with the results / score of another test which we think is a standard that we want to achieve, which is a standard that we want to compare with

Criterion- related validity refers to a procedure where scores o a test being used are correlated with scores on a criterion

The evidence that a test score corresponds to an accurate measure of interest. The measure of interest is called the criterion” (Kaplan & Saccuzzo, 2001, p.635).

Why do we need a criterion to estimate the validity of our measure? To understand this concept let’s look at two simple everyday example

In other words, for example, if a teacher develops a test of mathematical ability, how is she going to determine its validity? How will she judge that her test is measuring mathematical ability and nothing else? Most probably she will try to examine the relationship between scores on her test with another authentic test of the same ability, or may be a test of similar skills. By doing so, she will be assessing the criterion- related validity of her test.

Criterion Prediction Procedures:

□ Predictive validity evidence

“The evidence that a test forecasts score on the criterion at some future time” (Kaplan & Saccuzzo, 2001, p.638). Predictive validity pertains (related to) to prediction over a time interval. Predictive validation yields information that is most relevant to tests that are used for predicting future performance of an individual.

Predict over a time interval (latter sense)

There are situations where tests are used to classify, select, and even reject individuals.

For example tests are used for selecting students for an academic program, for personnel selection and job placement, or for choosing staff members for a specific training or higher education.

Similarly there are times where rather than being used for selection; tests are used for screening out (filter) individuals. The tests are used to predict if the candidate is likely to develop an undesirable behavior, characteristic, mental or emotional problem etc. in future.

□ **Concurrent validity evidence**

Concurrent validity is relevant to tests employed for diagnose existing status rather than prediction of the future

In this approach the test is administered to such a sample whose data on the criterion are already available

Concurrent valid evidence is defined as “evidence for criterion validity in which the test and the criterion are administered at the same point in time” (Kaplan & Saccuzzo, 2001, p.635).

When the validity of a test is measured with reference to a criterion, and both are administered at nearly the same time, then the resulting validity will be concurrent validity

The primary idea is whether this test can differentiate between people with specific characteristics. If the test can do so, then it can be used as an efficient tool for categorizing people according certain criteria or classifications. One of the best examples of such a tool can be diagnostic tests e.g. Minnesota Multiphasic Personality Inventory (MMPI).

The criterion data are always available at the time of testing.

“The logical distinction between predictive and concurrent validation is based not on time but on the objectives of testing. Concurrent validation is relevant to tests employed for diagnosis of existing status rather than prediction of future outcomes

Example

Does smith qualify as a satisfactory pilot? (Concurrent validity)

Does smith have the qualities to become a satisfactory pilot? (Predictive validity)

Criterion Measures

There is no limit to the type of criterion that may be used for the purpose of test validation. However some are used more frequently and popularly than others. Tests may involve a number of criteria. Some of the criteria are mentioned here:

- a) Academic achievement; used for intellectual tests and similar tests
- b) Matriculation marks at the time of admission to higher classes; used for selection and screening.
- c) For those who have completed, or discontinued education, years of education can be good criteria.

- d) Performance in specialized training; used for selection, admission, achievement. e) Instructor's ratings
- f) Grades in a semester

Characteristics of a Criterion:

Reliable; the scores on the criterion as well as those on the test should be reliable. The reliability coefficients of the two sets of scores affect the coefficient of validity. This can be understood from the following pattern of relationship

$$r_{xy} \leq \sqrt{(r_{xx})(r_{yy})}$$

r_{xy} stands for the coefficient of validity

Test reliability and criterion reliability are represented by r_{xx} and r_{yy} . It can be seen that the two coefficients of reliability contribute to the validity coefficient. Weak/little/or no reliability will automatically affect validity.

Valid; it should measure what it is supposed to measure.

Relevant; it should be relevant to the purpose for which it is being used.

Uncontaminated; (pollute) we should try to use uncontaminated criterion. There are situations when the criterion itself has been, partially or fully, based on predictor measures. This is called criterion contamination.

They should have similar things, similar trait, similar characteristics they may be based upon similar ideas but they should be independent of each other at the same time (same items)

Construct Validity

Construct validation has focused attention on the role of theory in test construction and on the need of formulation hypotheses that can be proved or disapproved in the validation process.

Some years back I developed a test for the assessment of science concept acquisition, called SCAT. The purpose of developing this test was to develop a tool that could measure the extent to which children in the final year of their school have learnt the application of science concepts that they were taught in the past 4-5 years.

The reliability and validity of the test were also measured. In order to see if the test was valid or not, three assumptions were formulated:

- The SCAT scores will have a high-positive correlation with scores on a test of similar nature and content. (similar nature)

- The SCAT scores will have a high-positive correlation with scores on a test of reasoning ability.(similar ability)
- The SCAT scores will have a moderate-positive correlation with marks on school science achievement test.
- The SCAT scores will have either no or a negative correlation with scores on a test of rote memorization of meaningless material like non-sense syllables

In order to test these four assumptions, four different tests were used for these assumptions respectively:

- Dallas Times Herald Test (scale test) (science test)
- Basic Reasoning skills Test
- General Science Test used in school's annual examination,
- list of non-sense syllable

Construct Validity Evidence:

“Construct validity is a judgment about the appropriateness of inferences drawn from test scores regarding individual standings on a variable called a ‘construct’. A construct is an informed scientific idea developed or constructed to describe or explain behavior”.

“Constructs are unobservable, presupposed (underlying) traits that a test developer may invoke to describe test behavior or criterion performance” (Cohen, 1999) **The significant elements of this description are as follows: (steps)**

- A theoretical construct or trait.
- Each construct is developed to explain and organize observed response consistencies.
- It derives from established interrelationships among behavioral measures. (tool)
- Requires the gradual accumulation of information from a variety of sources

Assumptions Underlying Construct Validity:

- Test scores will be highly correlated with tests measuring same construct/similar construct. This is **convergent validity**.
- Test scores will have weak/low (or at times may be negative) correlation with tests meant to measure construct which are different from the one measured by the main test. This is **discriminant validity**. (covers a different construct)

Convergent Validity Evidence: This aspect of validity shows us the extent to which a test measures the same construct/trait/ attribute as do other measures that have been designed and developed to measure the same construct/trait/ attribute

Discriminant Validity Evidence: Discriminant validity provides us information that the test in question does not measure what other tests measure, or it measures something different from what other available tests measure

Can we measure both validity aside? Because

they support each other validity

Steps in Construct Validity Process:

- The specific test being referred to must be present.
- Clearly defining the construct to be examined
- Assumptions about the construct
- Assumptions about the similar constructs and the different constructs
- Identification of measures (tests) that will be used to test the constructs in question
- Test administration, may be in various steps (session , situation)
- Calculation of correlations between the test scores
- Analysis of correlations in order to reach final conclusions regarding construct validity

Factors Affecting Test Validity

a) Nature of the Group

- Tests are validated on specific groups, and results may vary with different populations.
- Test manuals should provide details of the validation sample for clarity.

b) Heterogeneity of the Sample

- Validity coefficients may be higher in diverse groups than in homogeneous ones.
- Consider the range of scores and diversity when interpreting validity.
- c) Preselection
- Validity in job selection tests may be affected if the sample is preselected.
- Preselected groups may limit the score range, influencing the validity coefficient.

d) Form of Relationship Between Test and Criterion

- Validity coefficients assume a linear relationship; deviations may impact accuracy.
- Ensure the relationship between the test and criterion is uniform across score ranges.

Standard Error of Estimate (SEest)

The concept of validity and its magnitude entail another concept i.e., the amount of error involved. This is the amount of error that may be involved in decisions based on a test whose validity is not

perfect It is described as “The error of estimate shows the margin of error to be expected in the individual’s predicted criterion score, as a result of the imperfect validity of the test”(Anastasi & Urbina,2007, p.157). The formula for SEest is as follows: $r_{xy^2} = \text{validity coefficient squared SDy}$ = the standard deviation of the criterion score

Test Norms

Definition and types of Norms.

Raw test scores need to be turned into a more meaningful form. Therefore they are converted into relative measures or derived scores. These norms help to compare an individual's test scores to the scores of others in the same group, providing a basis for understanding how well someone performed relative to their peers. Norms are essential for interpreting test results and determining where an individual stands in comparison to the larger group that took the same test.

Norms performs two functions

Standard and comparative function

Standard means within a test if there’s many subtest we can compare the performance of various subtest with each other

Comparative means that we can compare our performance with the norms performance.

Developmental Norms

Test scores can be described using developmental norms. Developmental norms refer to the usual patterns, characteristics, and age-specific tasks or skills associated with development at any age or stage

The underlying assumption is that people, children and adults, are capable of performing at specific levels at different stages of life

When most people can perform certain tasks at a certain age level then it is considered as the norm for that age level

Subsequently if a 13 year old girl can perform the tasks accurately and completes a test that most 13 year olds can do, then her mental age will be stated as 13 (MA = 13)

If an adult can perform only the tasks that a six year old can do, then his MA will be 6. Considering his physical age, he is mentally deficient, backward, or special.

Developmental norms may also be based upon highly qualitative descriptions of behavior in specific functions such as sensorimotor activities or concept formation, expressed in qualitative terms

Difficulty in developmental norms

But this approach is not as easy to apply as it may seem to be. At times people take tests that measure different abilities. Even the subtests of the same test may be measuring a variety of skills or abilities. In such cases it is not necessary that everybody attains the same MA in all tests or subtests. This makes comparison in terms of developmental norms difficult.

Example

A language proficiency test with subtests for speaking, listening, reading, and writing. If a person performs exceptionally well in speaking and listening but struggles in writing, their overall language proficiency score will reflect a mix of these different language skills. This variation in performance across various subtests makes it challenging to compare individuals using a single developmental norm, as people may exhibit different levels of proficiency in distinct aspects of language.

Another concept that use in the reference in developmental norm that is Mental Age

□ Mental age

The use of this terminology began after Simon and Binet scale introduced he himself never use this term Mental age he prefer to use the term Mental Level but other psychologist who use this test prefer to use Mental age

In Binet's scale items were grouped in year levels e.g items that were passed by the majority of **8 years old** children in the standardization sample were included in the **8- year level**. Similarly items that were passed by a large number of **10 year old** children were placed in **10- year level** and so on. Stanford- Binet and other similar scales are age scales

While the scale was being used very frequently, the problem of '**scatter**'(**different score in different subtest**) of scores was observed. It meant that many subjects did **not show uniform performance** on all subtests of the scale/test. Some individuals would failed the tests that were **below their age** level while at the same time they had passed the tests that **were above their age levels**. For overcoming this problem the concept of "**basal age**" was introduced.

Basel age is the highest age at which a person successfully completes all the test items. If someone passes some items at older age levels, they receive partial credit in months for those particular items.

The partial credits in months, along with the basal age, were combined to calculate the child's mental age.

Example

Imagine a child taking a math test. The basal age is determined as the highest age level at which the child successfully completes all the initial (easier) math problems. Let's say the child passed all the items up to the 7-year-old level.

Now, suppose the child attempted some harder problems meant for 8-year-olds and 9-year-olds. The child got partial credit for these; say 3 additional months for the 8-year-old problems and 2 additional months for the 9-year-old problems.

To find the child's mental age, you add the basal age (7 years) to the additional months of credit (3 months + 2 months), resulting in a mental age of 7 years and 5 months. This combined value represents the child's overall performance on the test, taking into account both the basal age and the credit for items passed at higher age levels.

Shortcoming

Mental age does not mean the same thing at different stages of life

MA of 4 of a 5 year old is not the same as the MA 24 of a 25 year old. As age progresses the unit of MA tends to shrink. The backwardness increase time to time

Therefore positive or negative deviation from norm at different age groups does not mean the same. Deviation at a very young age means a lot as compared to that at older age.

□ Grade Equivalents:

Grade equivalents represent the scores on educational achievement tests attained by children in a certain grade.

If 6th-grade children, on average, scored 35 on an arithmetic test, a student scoring 35 is said to have a grade equivalent of 6, indicating performance at the typical level of a 6th grader.

Another problem is Grades usually last for about 10 months 2 months are vacation (summer , winter) if a child is tested when he just entered grade 7 is he going to perform the same as a child who has been in grade 7 for the past six months, In such cases the successive months can be expressed in decimal points

Example

The grade equivalent of 7.0 refers to the average performance of a 7th grader at the beginning of session. 7.5 present the average at the middle of the session and so on.

Shortcomings

Teacher factor course factor content factors etc.

- **Ordinal Scales:**

Another approach to developmental norms develops from the research in the area of child psychology.

Ordinal scales describe phenomenon /event/ object, in certain rank order

Example Psychologists studying infants and young children observed changes in their abilities over time. They documented functions such as sensory discrimination, language development, and concept formation at different ages. These practical observations have proven valuable in enhancing our understanding of human development.

There are **two important** things in developmental level

- Developmental levels are sequential
- They are uniform

Within-Group Norms

Within-group norms are like comparing how well someone does to a similar group of people.

For example, when a child takes a test, their score is compared to other children of the same age or grade. This way of scoring is common, and now most tests use it in some way. The scores are calculated using specific mathematical methods, making them easy to understand in numbers.

Percentiles

A percentile shows where a person stands compared to others who took the same test.

(Comparison of your score with other group) rank

It tells you the percentage of people who scored lower than you.

- Percentiles can also describe ranks in a group of 100 people i.e a person who is at the top in the group given the rank of one; likewise a person who is at the bottom in a group will be given a poorer rank.
- The difference between percentage and percentile is that percentage is a raw score while percentiles are derived scores.

When we have to compare the score with one test to another or one person with another at time use of percentile does not serve the purpose we need some other concept to compare

Standard Scores

Standard score is how we derive raw score in this way that it will become comparable

Raw score of a person **with reference** to the average (mean), of the norm and the standard deviation of the norm.

Known as z score

Computation of Standard Scores

$$z = \frac{X - M}{SD}$$

X: Raw score

M: test score (mean score)

Z: standard score

Normal Standard Scores

It gives us information about relative position of a person or of a score in given distribution. A standardized test score is usually represented as a number indicating how well a child performed on an assessment.

The Deviation IQ:

The term IQ (Intelligence Quotient) was introduced in early intelligence tests. It is simply obtained by dividing the MA by chronological age, and multiplied by 100:

$$IQ = MA / CA \times 100$$

The Ratio IQ has been found to have significant technical issues. One problem is that IQ scores are not easily comparable across different age groups with different tests.

It compares people of the same age and assumes that IQ of individuals is normally distributed

Scales of Measurement:

To describe test scores in numbers, the tests need to provide results in a numeric form. This means the answers should be numbers or be able to be changed into numbers. When measuring things like personality, rules are used to give numbers to things and express qualities as numbers. For example, in a personality test, a question might ask, "Do you like to be with young people a lot?" The answers are given in words like "always," "often," "could not say," "rarely," and "never." The person taking the test has to pick the one that best describes them. This response is in words, so it can't be compared with others. To make it possible to compare, each word is given a number. For example, "never" might be 1, and "always" might be 5. Now all the person's answers can be turned

into numbers, which can be used for statistics. In short tests where there's a right answer for every question, like in ability tests, the number of correct answers is counted to get the test score.

Test Norms and Related Concepts

Standardization

Norms are set to standardize a test, meaning it's given to a group representing the population it's intended for, to establish average scores. A standardized test includes normative data and clear instructions for administering and scoring.

Example

Imagine you're creating a new math test for high school students. To standardize it, you give the test to a group of high school students from different schools to see how they perform. The average scores from this group become the norm or standard.

Now, when someone else takes the test, you compare their score to this average to see how well they did compare to the typical high school student. A standardized test, in this case, includes information about what's considered a typical score for high school students and clear instructions on how to give and score the test consistently.

The Normative Sample

A normative sample refers to a group of individuals who are representative of the larger population and are used as a standard for comparison in testing or research. The performance or characteristics of this sample help establish norms or average values against which individual scores or behaviors can be compared.

- **Representation:** Ensure that the sample includes people from different regions, communities, socioeconomic backgrounds, and relevant demographics to accurately represent the entire population of interest.
- **Random Selection:** Use random and unbiased methods when selecting individuals for the sample to avoid any unintentional skewing of results.
- **Sample Size:** Ensure the sample size is large enough to provide reliable and valid statistical results. A larger sample size generally increases the precision of the norms.
- **Inclusivity:** Make efforts to include diverse groups within the sample to account for variations in characteristics that may impact test performance.
- **Relevance:** Ensure that the sample is relevant to the purpose of the test. For example, if the test is designed for university students, the normative sample should consist of individuals who accurately represent the university student population.

- **Current Data:** Regularly update the normative sample to account for changes in demographics, education systems, or other relevant factors over time.
- **Ethical Considerations:** Follow ethical guidelines and obtain informed consent from participants to ensure the ethical treatment of individuals within the normative Sample.

National Norms

If a test is made standard using people from all across a country to represent everyone, it's called a "national sample." This group is chosen to cover various areas, communities, and backgrounds. For instance, if we want to set standards for a test for university students in Pakistan, we need to pick a group of university students that represents all parts of the country.

National Anchor Norms

When measuring the same skill using different tests, people might get different scores. Psychologists need to interpret these scores and understand how a score on one test compares to a score on another. National anchor norms help solve this by offering a table that shows the equivalence of scores from different tests measuring the same skill. The equivalence is determined using the 'equipercentile method,' where scores on different tests are compared based on their corresponding percentile scores.

Example

Imagine you take two different math tests, Test A and Test B. You score 80 on Test A and 90 on Test B. To understand how well you did in comparison to others, psychologists use a table based on national anchor norms.

The equipercentile method helps them see that your score of 80 on Test A is similar to the performance of someone at the 70th percentile, while your score of 90 on Test B is similar to the performance of someone at the 85th percentile.

So, even though the scores are different, the equipercentile method and the national anchor norms allow psychologists to understand how your performance on one test compares to your performance on another. This way, they can interpret and compare scores across different tests more accurately.

Specific norms

Specific norms are benchmarks or standards that are customized for a particular group, context, or population. They are designed to provide a targeted evaluation within a specific setting, taking into account the unique characteristics and demographics of that particular group. These norms offer a more detailed and context-specific assessment compared to general norms. **Example**

Imagine a middle school introduces a new way of teaching math to its eighth-grade students. Instead of comparing their progress to general standards that apply to all schools, they create

specific norms based on how well their own eighth-grade students are doing. They look at things like the students' previous math knowledge, the effectiveness of the new teaching methods, and any unique challenges their students might face.

So, the specific norms they create help the school understand how successful their students are in learning math with this new approach, considering the specific factors that are relevant to their school environment.

Local norms

Sometimes, instead of using general norms or even specific norms, organizations like universities or companies create their very own norms. These are called local norms. **For example**, a university might collect information about how well first-year students perform and use that data to predict how they will do in the coming years. **Similarly**, a company might set its own norms for new employees and use it to guess how well they'll perform in the future. In both cases, they gather information about performance and progress to make these predictions.

Fixed Reference Group

In the context of testing and scoring, a fixed reference group refers to a specific group of individuals whose scores on a test are used as a constant benchmark for future test administrations. The scores obtained by this initial group become the reference point against which the scores of subsequent test-takers are compared.

For example, if a company uses a fixed reference group scoring system for an employment assessment, the scores obtained by the first set of employees who took the test might serve as the reference. Any future scores are then interpreted in relation to how they compare to the performance of this initial group. This approach is an alternative to using broader, normative samples for comparison.

Item Response Theory (IRT)

Item Response Theory (IRT) is like a smart way of understanding test questions. Back in the 1970s, when fast computers became available, psychologists started using advanced math to figure out how hard test questions are. They called these methods "latent trait models."

These models measure the chance that someone with a certain ability will get a question right. They don't say there's a specific trait; it's more like a statistical guess based on observed test responses. Your total score on a test is just a rough estimate of your underlying ability.

Later on, they replaced "latent trait model" with Item Response Theory (IRT) because "latent trait" gave the wrong idea about having a specific trait.

The goal of IRT is to create a fair way of measuring ability that works for everyone, no matter how good or not-so-good they are at the subject. Instead of using the average and spread of scores from

one group to set the scale, IRT uses data from many different groups to define the scale in a more flexible way.

Norms Referenced test

To develop a norm-referenced test, the test developers select a statistically relevant group of individuals and administer the test items. The scores of this norming group are used to create the scoring system for that test. The composition of the norming group depends on the test, but factors considered usually include age or grade level, and may also be narrowed down by other demographic information. In addition, some tests are normed for more than one group. If so, a test administrator gives the test, then chooses the correct scoring system based on the subject's qualifications (i.e., the administrator uses a different scoring chart for a seven-year-old than for a twelve-year-old).

The norm-referenced test definition indicates that the results are reported as a percentage or a percentile ranking. The purpose of this number is to tell the test-taker what percentage of the norming group scored above and below them.

A criterion-referenced assessment

A criterion-referenced assessment is a type of test or evaluation in which a person's performance is measured against predetermined criteria or standards, rather than being compared to the performance of others, as is the case with norm-referenced assessments.

Key characteristics of criterion-referenced assessments include:

1. **Objective Standards:** Criterion-referenced assessments are designed with specific, predetermined criteria or standards that define what individuals are expected to know or be able to do. These standards are often based on educational or performance objectives.
2. **Individual Mastery:** The focus of criterion-referenced assessments is on determining whether an individual has mastered a particular set of skills or knowledge. The goal is to assess how well a person can perform specific tasks or meet established criteria.
3. **Absolute Measurement:** Instead of comparing individuals to each other, the assessment measures their performance against an absolute set of standards. This makes criterion-referenced assessments particularly useful for determining whether someone has achieved a certain level of proficiency or competency.
4. **Feedback for Improvement:** Criterion-referenced assessments often provide detailed feedback to individuals, educators, or organizations about which specific skills or content areas need improvement. This information can be valuable for guiding further instruction or training.

Example

A psychological test designed to assess how severe someone's depression is. The test has specific criteria for different levels of depression:

- Minimal Depression: Scores between 0-10
- Mild Depression: Scores between 11-20
- Moderate Depression: Scores between 21-30
- Severe Depression: Scores above 30

Now, if an individual takes this test and scores 15, the interpretation is not based on how that person compares to others. Instead, the focus is on the predefined criteria. In this case, a score of 15 falls within the range associated with mild depression.

Test Construction and Standardization

Different taxonomies of test Development

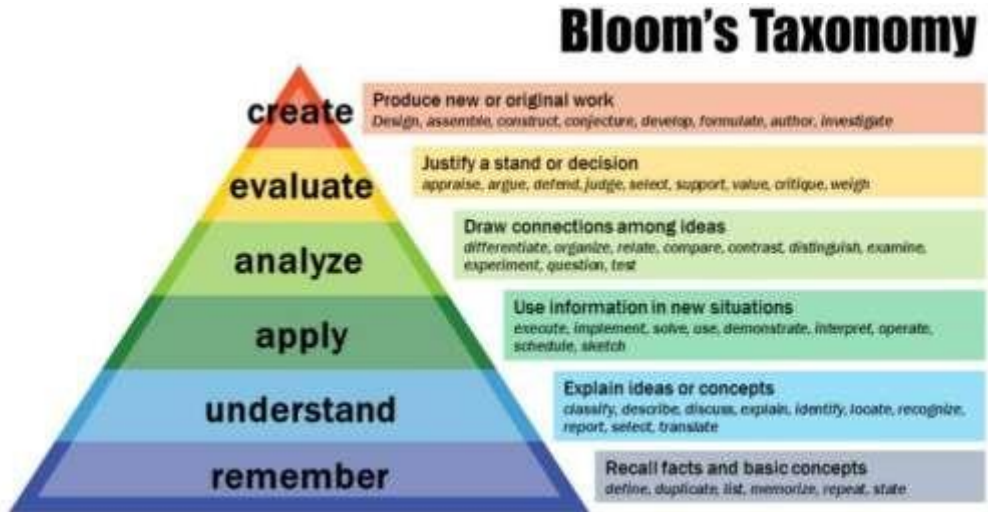
Taxonomies refer to systematic classifications or categorizations of the various components or aspects involved in the process. Different taxonomies provide a structured framework for understanding and organizing the elements of test development. Here are a couple of common taxonomies related to test development:

1. Bloom's Taxonomy:

- Bloom's Taxonomy is a well-known classification system in education that categorizes educational objectives and skills into hierarchical levels. It was originally developed by Benjamin Bloom and his colleagues in the 1950s. The taxonomy is often represented as a pyramid with six levels, starting from the basic cognitive skills at the bottom to higher-order thinking skills at the top. The levels include:
 1. Knowledge
 2. Comprehension
 3. Application
 4. Analysis
 5. Synthesis

6. Evaluation

Bloom's Taxonomy is often used to guide the development of assessment items and to ensure that a range of cognitive skills is targeted in educational assessments.



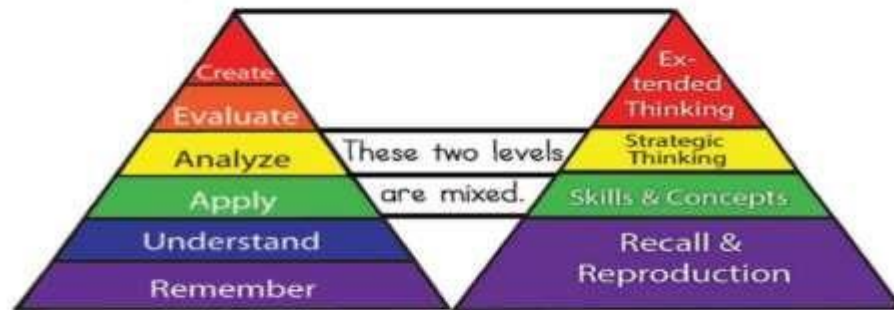
2. Webb's Depth of Knowledge (DOK):

- Developed by Norman L. Webb, the DOK taxonomy categorizes content complexity in terms of the cognitive demand required to complete a task. It classifies tasks into four levels:
 1. Level 1: Recall and Reproduction
 2. Level 2: Skills and Concepts
 3. Level 3: Strategic Thinking
 4. Level 4: Extended Thinking

Webb's DOK is often used in educational settings to design assessments that go beyond simple recall and encourage students to engage in more complex thinking processes.

Bloom's Vs. DOK?

This is an over-simplification of the correlations, but it is SOMETIMES works!



DOK needs more complexity at each higher level. Generally for primary children, if you can get kids to think about more than one text at a time and get them to form opinions AND support them, you'll hit level three in DOK. The Extended Thinking level in DOK needs multiple texts, complexity, and a project completed over time, such as in Project Based Learning Activities.

3. Item Taxonomies:

- In the context of test development, item taxonomies categorize test items based on the type of response they require. For example:
- **Selected Response Items:** Multiple-choice, true/false, matching items.
- **Constructed Response Items:** Short-answer questions, essays, performance tasks.

4. Content Taxonomies:

- These taxonomies classify the content or subject matter of the test. For instance, in a science test, the taxonomy might categorize questions into topics such as physics, chemistry, biology, etc.

5. Cognitive Process Taxonomies:

- These taxonomies focus on the cognitive processes involved in answering test questions. They may include categories like recall, analysis, synthesis, and application.

These taxonomies help guide the development of assessments by providing a structure for defining the goals of the test, specifying the cognitive demands of the items, and organizing the content being assessed. They are valuable tools for educators and test developers to ensure that assessments are comprehensive, aligned with learning objectives, and promote a range of cognitive skills.

1. Test Conceptualization

Test development is a result of test developers' idea of developing a tool to measure a particular construct. The stimulus for developing a test can be anything.

For example, Literature on an already developed test might create the need for further work on the psychometric soundness of the test, and the would-be test developer thinks that he/she can do better.

The emergence to prominence of some social phenomenon or pattern of behavior might serve as the stimulus for development of a new test. Apart from the stimulus for developing a new test, a number of questions immediately confront the prospective test developer. Some of these questions include

- What is the test designed to measure?
- What is the purpose of developing the test?
- Is there any need for this test?
- What will be the sample of the test?
- What should be the test content?
- What should be the procedure for test administration?
- What should the ideal format of the test be?
- Should more than one forms of test be developed?
- What special training will be required of test users for its administration and interpretation?
- What type of responses will be required by test takers
- Who will get benefit from its administration
- Is there any potential for harm as the result of an administration of the test? □ How will meaning be attributed to scores on test?

The last question points out the issue of norm versus criterion referenced tests. There are different approaches to test development depending upon whether they are criterion referenced or norm referenced tests. A good item on a norm-referenced achievement test is an item for which high scorers on the test respond correctly; low scorers on the test tend to get that very same item incorrectly. Whereas, development of a criterion oriented test or technique entails pilot work with at least two groups of test takers' one group known to have mastered the knowledge or skill being measured and another group known to have not mastered such knowledge or skill. The items that best discriminates between these two groups would be considered "good" items

Pilot Work

In the context of test development, pilot study/research refers to preliminary research surrounding the creation of a prototype of the test. Test items may be pilot studied to evaluate whether they should be included in the final form of the instrument. In pilot work, the test developer typically

attempts how to best measure the targeted construct. The process may involve the creation, revision, and deletion of many test items. Once pilot work has been completed, the process of test construction begins. However, in future the need for pilot research is always a possibility because of the test's requirement for updates and revisions.

Scaling

Scaling is a crucial step in creating a test, where we establish rules for assigning numbers to measure different aspects. In simpler terms, it's like deciding how to give values to the various things we want to assess.

For example:

Example: Classroom Participation Score Scaling

Suppose a teacher wants to assess students' participation in class discussions and activities. The teacher decides to create a scaling system to assign scores based on participation levels.

1. Raw Data:

- Students participate in class discussions, and the teacher observes their engagement. The raw data could be the number of times each student actively contributes in a week.

2. Scaling Rules:

- The teacher establishes scaling rules to convert the raw participation data into scores. For simplicity, let's say the scaling ranges from 1 to 10.

3. Assigning Scaled Scores:

- The student who participates the most frequently might receive a scaled score of 10, indicating excellent participation.
- A student who participates moderately might receive a scaled score of 5, representing average participation.
- A student who rarely contributes might receive a scaled score of 1, indicating minimal participation.

4. Interpretation:

- The scaled scores now provide a standardized way to understand and compare students' participation levels. A higher score implies more active involvement in class discussions.

In this scenario, scaling helps the teacher quantify and evaluate a somewhat subjective measure (classroom participation) by assigning numerical values based on established rules. It simplifies the process of assessing and comparing students' participation, making the evaluation more objective and consistent.

Types of scales:

Scales help us measure things in different ways. While we often talk about nominal, ordinal, interval, and ratio scales based on the level of detail, we can also categorize scales in other practical ways.

- **Age Scale**
- **Grade Scale □ Stanine Scale**

• Age scale:

If we're particularly interested in understanding how people of different ages perform on a test, we might label it as an age scale. This means that the test helps us see how well individuals from various age groups do in comparison.

Example: Imagine we're creating a test to assess cognitive development in children. If our main interest is to understand how different age groups of children perform on the test, we might call it an age scale. This means the test helps us see how cognitive abilities vary across age ranges, such as 5-year-olds, 8-year-olds, and so on.

- **Grade scale:** When we want to know how students from different grades perform on a test, we call it a grade scale. This kind of test helps us see if there are patterns or differences in performance based on the students' grades.

Example: Suppose we're developing a math test for students. If we want to analyze how well students from different grades (like 5th, 8th, and 10th grades) perform, we could refer to it as a grade scale. This allows us to explore if there are grade-level patterns in math proficiency.

- **Stanine scale:** A Stanine, short for "standard nine," is a method of scaling test scores on a ninepoint standard scale. The stanine scale is often used in educational assessments to simplify the interpretation of test results.

Example: Consider a standardized test with raw scores. If we decide to convert these raw scores into stanine scores ranging from 1 to 9, we are using a stanine scale. For instance, a student who scored in the 8th stanine would be interpreted as having a high level of performance compared to others who took the test.

Scaling Method

Scaling methods help us measure things like attitudes or preferences. One way to do this is with a **Likert scale**. It's simple – we ask people questions, and they choose from five responses, like "agree" to "disagree" or "approve" to "disapprove." Likert found that giving scores of 1 through 5 usually works well.

Likert Scale Example:

- **Question:** "How satisfied are you with the new product?"
- **Likert Scale Responses:**
 - 1 - Very Dissatisfied
 - 2 - Dissatisfied
 - 3 - Neutral
 - 4 - Satisfied
 - 5 - Very Satisfied
- **Response:** The respondent selects the number that best represents their level of satisfaction.

Another method is **paired comparisons**. We show people pairs of things and ask them to pick the one they like more. They get a higher score if they choose the option that most people agree is better.

Paired Comparison Example:

- **Task:** Show respondents pairs of two different smartphone models (Model A and Model B) and ask them to choose the one they find more appealing.
- **Pair :** Model A vs. Model B
- If the respondent chooses Model A, it gets a point.
- **Overall Result:** Based on the choices, we can determine which smartphone model is more preferred overall.

Sorting tasks are another way. We might show people cards, pictures, or objects and ask them to put them in order. One type is comparative scaling, where you judge each thing compared to everything else. This helps us understand the order or importance of each item.

- **Task:** Provide a set of pictures of animals and ask respondents to sort them based on perceived intelligence, from the most intelligent to the least.

- **Comparative Scaling:** The respondent judges each animal compared to every other animal in terms of intelligence.
- **Result:** By analyzing the sorted order, we gain insights into which animals are generally perceived as more intelligent by the group.

Categorical scaling

Categorical scaling is a way of organizing things by sorting them into different groups. These groups have differences that can be measured. For example, we might put items into categories like "small," "medium," and "large."

Example: Categorical scaling is like sorting things into groups based on their qualities. Let's say we want to sort fruits by their sweetness:

1. **Category 1: Very Sweet (e.g., Mango)**
2. **Category 2: Moderately Sweet (e.g., Apple)**
3. **Category 3: Less Sweet (e.g., Lemon)**

All the methods mentioned give us information that is in order but not exact. Another method called the method of **equal-appearing intervals**, introduced by Thurstone, is used to get data that we think is more exact, like measuring things on a scale.

Method of equal-appearing intervals helps us measure sweetness more precisely. Imagine we have a scale where each step is equally sweet:

- Equal-Appearing Interval Scale:
- 1 - Not Sweet at All
- 2 - Slightly Sweet
- 3 - Moderately Sweet
- 4 - Quite Sweet
- 5 - Extremely Sweet

Now, if we place each fruit on this scale, we can better compare and measure their sweetness. For example, Mango might be a 5, Apple a 3, and Lemon a 1. This way, we're not just sorting them into categories; we're also giving a more detailed measure of their sweetness.

Writing Items:

The process of test construction also involves the ideas related to item writing. The three important considerations in this regard are:

- What range of content should the items cover?
- Which of many different types of item formats should be employed?
- How many items should be written?

When making a standardized test with multiple-choice questions, it's a good idea to start with twice the number of questions needed for the final test. This is because some questions might be removed later to make the test better. The final test should still cover all the important topics.

The person creating the test might write many questions based on their own knowledge. They can also get help from experts or learn from the group of people who will take the test. Literature searches may also be a valuable source of inquiry for item writing. Looking into existing information, like books or articles, can also be useful.

Decisions about the test, like its purpose and how many people will take it, depend on different things. A good way to ask questions is to give many possible answers. This is called a **selected response format**, and it helps participants choose the best answer from a list of choices.

Example of selected response format

Which of the following defense mechanisms is characterized by attributing one's own unacceptable thoughts or feelings to others?

Options: a. Repression b. Projection c. Rationalization d. Sublimation

Correct Answer: b. Projection

Test-takers choose the most appropriate option from the provided list (options a, b, c, and d).

Item Formats:

There are two types of format; selected response format, and constructed response format.

Selected response format:

This type of format presents the examinee with a choice of answers and requires selection of one alternative. The types of selected-response format are multiple choice, matching, and true/false items.

Example: **Question:** What is the capital city of France?

Options: a. Berlin b. Paris c. Rome d. Madrid

Answer: b. Paris

Constructed Response Format:

In this format, you have to provide or create the correct answer instead of just choosing it. There are three types of constructed-response items:

- A completion item asks you to give a word or phrase to complete sentences.
- A good short answer item is clear, and you can respond briefly with a short answer.
- An essay is a format where you describe a single topic in detail as asked in the question.

Examples:

Completion Item:

Task: Complete the following sentence: "The sun rises in the _____."

Answer: East

Short Answer Item:

Question: What is the capital of Pakistan?

Answer: Islamabad

Essay:

Topic: Describe the impact of technology on modern education.

Answer: In the past few decades, technology has revolutionized education by providing new tools for learning, facilitating online resources, and enhancing communication (continue).....

Scoring Items:

There are different ways to score tests, but one common method is the **cumulative model**. In this model, if someone gets a higher score on the test, it means they likely have a higher ability or trait related to what the test is measuring. Each correct response a person gives adds up, showing their overall skill or characteristic.

Example of Cumulative Scoring: Johnny earns a score of 8 for his ability in math because he got 8 questions right. The higher the score, the better Johnny's math skills are perceived to be.

Another scoring model is the **class model**. Here, people's responses help place them in a group with others who have similar patterns of scores. This is useful for sorting test-takers into different categories based on their performance.

Example of Class Scoring: Participants who share similar communication styles are grouped together. For instance, those who scored high on assertiveness might be in the "Confident Communicators" category, forming a class of individuals with similar patterns of scores.

The third model is **ipsative scoring**. In this method, the focus is on comparing a person's scores on different parts of the same test. For example, how well they do on one section compared to another within the same test.

Example of Ipsative Scoring:

Question: In a career assessment, Sarah rates her preferences for working independently and in a team on a scale.

Ipsative Scoring: Sarah's scores are compared within the same test. If she rates herself high on working independently but lower on teamwork, it provides insights into her preferences within the context of the assessment.

Once all the groundwork is done, and a test draft is ready, the next step is to try out the test to see how well it works.

3. TEST TRYOUT

Having created a pool of items from which the final version of the test will be developed, the test developer next tries out the test. The test is tried out on the sample for which it is constructed. It is also important to consider that on how many subjects the test should be tried out. It is usually considered that no fewer than five subjects and preferably as many as ten subjects, for every one item on the test. The more the subjects in try out, the better it is. Test tryout should be executed in the same conditions that are same as possible to the conditions under which standardized test will be administered. Test instructions, and everything from the time limits allotted for completing the test, to the atmosphere at the test site should be similar as possible.

What is a Good Item?

Characteristics of a good test are considered to be characteristics of a good item.

1. A good test is one that is reliable and valid; similarly a good test item should be valid and reliable.
2. Further, a good test items helps to discriminate test takers; a good test item is one that high scorers on the test as a whole get right. An item that high scorers on the test as a whole do not get right is probably not a good item.

3. A good test item can also be described as one that low scorers on the test as a whole get wrong; an item that low scorers on the test as a whole get right may not be a good item.

After the first draft has been administered to a representative group of examinees, it remains for the test developer to analyze test scores and responses to individual items. At this stage the test undergo different types of statistical analyses which are collectively called as “item analysis”

4. ITEM ANALYSIS

Statistical procedures are collectively known as **item analysis**. For item analysis different statistical procedures are employed in order to select the best items from a pool of tryout items. Among the tools that test developer employs to analyze and select items is an index of an item's difficulty, an item-validity index, an item-reliability index, and an index of item discrimination.

Qualitative Items Analysis: Though statistical procedures are employed for item analysis, there are some non-quantitative methods that employ verbal rather than mathematical techniques. Through use of simple questionnaires or individual or group discussions with test takers, any test user can obtain valuable information on how the test could be improved.

5. TEST REVISION

During item analysis, a lot of information is collected. Using this info, some questions are removed from the original set, and others are rewritten. One approach is to assess each question's strengths and weaknesses. If many good questions are too easy, the developer might intentionally add some more challenging ones for balance. After addressing all concerns, the revised test is improved in quality. The next step is to give the updated test under standard conditions. By analyzing the data from this test, the developer considers it finished.

Types of Items

The process of test construction also involves careful planning, considering a number of factors to be kept in mind before actually writing test items. Three important considerations in this regard are:

- What range of content should the items cover?
- Which of many different types of item formats should be employed? □
How many items should be written?

Nature and Range of Content:

The test developer will have to decide about the nature and extent of content to be included in the test. This will primarily be decided with reference to the objectives laid down for measurement. Each item should be measuring some aspect of the content area.

Type of Items: Different types of formats are available to test developers. The type of format will also depend upon the type of test and construct or trait being measured. For example a projective item may be suitable for a personality test but not for an achievement test. Details of item formats are given in the following sections.

Number of Items:

The test developer also has to consider the number of items to be included in the test. It will affect the length of the test. Also, whether the test is going to be administered individually or in group may also be affected by this decision. Another variable that may affect decision about the number of items is whether the test is going to measure a single trait/ ability/ domain or multiple traits/abilities/domains. Fewer items will be required if a single domain is to be measured. In case of multiple traits/ abilities a larger number of items will be required so that there are enough items available to measure each one of them. We have very lengthy inventories like MMPI containing a few hundred items, and short scale such as self-efficacy scale containing 3 to 10 items. When a standardized test is developed which is based on multiple-choice response format, it is usually advisable that the number of items for the first draft of a standardized test contain approximately twice the number of items that the final version of the test will contain. The test developer may write a large number of items from personal experience. Help from experts can also be taken for item writing. Literature searches may also be a valuable source of inquiry for item writing. Considerations related to variables such as the purpose of the test and the number of examinees to be tested at one time, enter into decisions regarding the format of the test.

Item Formats:

Kaplan and Saccuzzo (2001) have given a very good description of test item formats. The test developer can choose from the following formats described by them:

- a) The dichotomous format
- b) The polytomous format
- c) The Likert format
- d) The category format
- e) Checklists and Q –sorts

Test formats are classified in many other ways as well; recall type and recognition type; constructed response type and identification type.

A very common distinction is made between objective type and essay type formats.

Another way is to divide test formats as selected response format, and constructed response format.

Selected Response Format: This type of format presents the examinee with a choice of answers and requires selection of one alternative e.g. on achievement test, the test taker is required to select the correct option. The types of selected-response format are multiple choice, matching, and

true/false items. Test formats described by Kaplan and Saccuzzo (2001) fall into this category. We will discuss these types in the following section. **A. The Dichotomous Format:**

Make two test items with alternate response options. One should be true/false type and the other one yes/no type

This is the alternate response format in which the test taker is provided with two response options to choose from. It can be used in a number of ways. In a common way,

one of the two options is right and the other one wrong. For choosing the right option a certain mark or score, usually one, is given e.g. ten right options marked will carry a score of ten. Another way is to use the 'yes' and 'no' options, as in many personality inventories. Here the subject tells if a statement describes him or not. An even more popular use is the true/false format. In this format the test takers are presented with a number of statements and they have to tell whether these are true or false. In case of teacher made tests, certain statements from the textbooks or other materials are used in these items. Items using these statements are 'true' whereas text materials are altered to develop items that are 'false'.

Such items have the following advantages:

- They are easy to develop.
- The test developer does not have to make effort to think of a number of options that seem to be correct.
- Development of such tests does not take too much time. □ These items are easy to score.

There are some disadvantages as well:

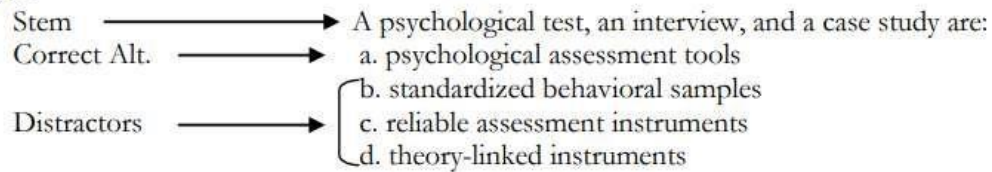
- A test taker may be able to accurately answer at least 50% answers even when he does not know the right answers. The probability of answering an item correctly by chance is $\frac{1}{2}$. If the teacher had used equal number of true and false items in the test and if the test taker marks all items as true (or marks all as false) then he will be marking half of the items correctly.
- Such items may encourage rote memorization of course content in case of teacher made tests. When students know that the items will be based on the text of the course content then they try to memorize the content rather than learning the concepts.

B. The Polytomous (Polychotomous) Format:

In this type of items multiple response options, rather than alternate response options are given. Such an item includes these elements:

- A stem
- A correct alternative option
- Several incorrect alternatives or options which are called "distracters" or "foils"

Example:



These response options include one right answer and the others are distractors.

The question or the item statement is called the stem. The stem should clearly state the question or the problem.

The distractors should be framed in such a manner that each one of them should appear to be the right answer

However there is still a **possibility** that someone may be able to answer a certain percentage of questions even without knowing a word of it. If the items in a test have four options each then there is a possibility that one may be able to answer **25%** of items correctly simply by making a right guess by chance. If three options are used then one may manage to mark **33%** of items correctly by chance

At times a correction for guessing is used employing the following **formula**:

$$\text{Corrected score} = R - W/n - 1$$

Where, R= the number of right responses
W= the number of wrong responses
n= the number of choices for each item
For example a test taker scores 40 out 100 in a test that used a four option format.

The score of 40 will be corrected like this:

$$\text{Corrected score} = 40 - 60/4 - 1 = 40 - 15 - 1 = 24$$

This example shows that one should understand that **guessing in a test may have more serious consequences** than one might foresee. If the scoring procedure involves correction for guessing then guessing is to be completely avoided. If one has little or no knowledge of the content then the likelihood of making wrong guesses will also be very high. Looking at the formula, it can be seen that the greater is the number of wrong answers the greater will be the number subtracted from the right answers, which means the smaller will be the corrected score.

Matching Item:

A matching item is the format different from multiple choice format.

The test taker is presented with two columns of responses and the test taker has to determine which item from the first column matches the item from the second column. Both columns may have equal number of items, hence making it easy for the examinee to match the items to which he does not know how to respond. E.g. if a respondent is unsure about one of the options give, he may deduce the right answer by matching all the other options first. Thus a perfect score would be obtained. Providing more options than are needed is designed to minimize such a possibility. C. **The Likert Format:**

A very popularly used tool for measuring personality and attitudes is the Likert's scale. Likert's scale format provides test takers an opportunity to endorse degree of their agreement to a statement. These statements pertain to attitude or personality. Likert used it as part of his method of attitude scale construction. Likert scales provide five response options ranging from 'strongly disagree' to 'strongly agree', with 'neutral' in between. For example: "I like to make friends who are older to me". Choose from the following options: Strongly disagree _____, disagree_____, neutral _____, agree_____, strongly agree_____ At times people have a tendency to mark 'neutral' rather than giving any clear cut answers. Therefore six options, rather than five may also be used:

Strongly disagree _____, moderately disagree_____, mildly disagree_____, mildly agree_____, moderately agree_____, strongly agree_____

The responses are summed to determine a person's score. In case of negatively worded items are reverse scored and added to the total

D. The Category Format

This format is similar to the Likert's scale, but has more options than five.

A 10-point scale is more commonly used. However the number of response categories may be more or less than that. This scale is used to rate something e.g. performance of a team. It is felt that people may not be very accurate in rating a player or a team's performance using this scale. It may happen that when a person watches a player in comparison to a superb player then he might rate the player to be rated at a lower level, whereas he may rate him very high when comparing with a poor performer E. **Checklists and Q-sorts:**

Adjective checklists contain a list of adjectives. The test taker checks the adjectives that are true about him. Checklists may be used for indicating one's own characteristics or those of other people. These are mostly used in personality assessment. In Q – sort the test taker is provided with a number of statements and has to sort them in nine piles depending on to what extent they stand true of her. This can be used for rating other people as well.

Constructed Response Format

It is the response format that requires the examinee to provide or create the correct answer than just selecting it. Three types of constructed-response items are the completion item, the short answer, and the essay.

- A completion item requires the examinee to provide a word or phrase that completes sentences.
- A good completion item should be clearly worded so that the correct answer is specified.
- A good short answer item is written clearly enough that the test taker can indeed respond briefly, with a short answer. There is no hard and fast rule that point out how short an answer should be.
- An essay is a type of response format in which the examinees are asked to describe in detail a single topic which is asked from them. The skills measured by essay type items are different from other type of item formats e.g. an essay requires recall, organization, planning, and writing ability, the other types of items require only recognition.

General guidelines For Item Writing

DeVellis (1991) has given a number of guidelines for writing items. Some of those are as follows

- **Clear definitions and item specificity:**

Whatever is to be measured should be clearly defined. Items should be as specific as possible.

- **Developing an item pool:**

Test developer should prepare a large number of items first and then choose items from this item pool. These items should all be representing the content area to be covered. Initially, the test developer may develop 3-4 items for each item to be included in the final version.

- **Appropriate level of reading difficulty:**

The level of test takers should be kept in mind while preparing test items. Test's reading difficulty level should be appropriate for their level. For example ' I have an inherent predisposition to procrastinate when my life gets stuck at crucial junctures and I have to take serious decisions to tackle and cope' includes many words that are not included in a layman's everyday vocabulary

- **Reasonable length of items:**

Items should not be too long. Exceptionally long items are hardly found to be good.

- **Avoid ‘double- barreled’ items:**

Double- barreled items are the ones that include two or more ideas at the same time. Practically speaking, these may be converted into two or more items depending on the number of ideas being conveyed. “I often help the poor because I believe in serving humanity and I am a follower of my leader who himself is known for social service.” This item conveys a number of ideas; “I often help the poor because I believe in serving humanity and I am a follower of my leader who himself is known for social service.”

- **Mixing of negatively and positively worded items:**

At times the test takers have a tendency to agree with most items. This is called ‘**acquiescence response set**’.

In order to overcome this problem positively and negatively worded items should be mixed and used alternately.

If all items are negative e.g. ‘I hate liars’, ‘Most people betray’, ‘I feel depressed most of the time’ then there is a chance that the test taker will develop a response set and will mark every statement as true, or may be false.

Alternately, positively directioned items may be added to the sequence in order to avoid the possibility of a response set. There are some other points that also need to be considered in test planning and development,

for example: Avoid double negatives: e.g. ‘I do not disagree to the fact that people should not be stopped from playing cricket’, or ‘Don’t you think that unsupervised internet use by children is not a good thing’. It is difficult for the test taker to understand what is being asked. These could have been rephrased as: ‘I agree to the fact that people should not be stopped from playing cricket’, and ‘Do you think that unsupervised internet use by children is not a good thing’ or even better ‘Do you think that internet use by children should be supervised?’ Besides this, the cultural background of would be test takers should also be kept in mind. Cultural, racial, and gender bias should be completely avoided.

- **Initial Test Plan and Design:**

From whatever we have discussed so far, we understand that development of any test involves the following steps:

- Determining and formulating the observations of the test

Deciding the domain and content to be covered ☐

Deciding about the format of the test and the test items ☐

Developing/ writing the items.

- Developing much more items than required
- Trying out the initial version
- Analyzing the results of the try out
- Reshaping and refining the first version
- Another try out if needed
- Norm development, if required (required in case of standardized tests) **Some Cautions:**
 - Avoid jargon. Use vocabulary that most people can understand
 - The test should not be too long
 - It should not be too easy that everyone can do it. Nor should it be that difficult that no one can do it.
 - Cultural biases and stereotypical ideas should be avoided
 - If any cultural differences are suspected to be present that may affect test results then nonverbal and culture free tests should be used.

But remember that even culture free tests may not be fully culture fair. People who have previous experience and exposure to such items may outperform those from such cultures where people are not familiar with tasks involving drawings and images.

Item Analysis

Item Analysis is “A set of methods used to evaluate test items. The most common techniques involve assessment of item difficulty and item discriminability”. (Kaplan & Saccuzzo, 2001)

Item writing refers to the process of creating test or assessment items, such as multiple-choice questions, true/false statements, or essay prompts. The goal is to design items that effectively measure the knowledge, skills, or abilities of the individuals being assessed. Here are several approaches to item writing:

1. Clarity and Conciseness

Write items that are clear, concise, and free from ambiguity.

Avoid using complex language or jargon that could confuse test-takers.

2. Alignment with Learning Objectives

Ensure that each item aligns with the learning objectives or outcomes of the assessment.

Items should reflect the content that was taught or the skills that were emphasized.

3. Relevance to Real-world Context

Frame questions in a way that relates to real-world situations or applications.

This helps test-takers see the practical importance of the knowledge or skills being assessed.

4. Distractor Quality in Multiple-Choice Items

Craft plausible distractors (incorrect answer choices) that are plausible and reflective of common misconceptions.

Avoid using distractors that are obviously incorrect or unrelated to the content.

5. Balance of Difficulty

Include items that vary in difficulty to effectively differentiate between high and low achievers.

Use a mix of easy, moderate, and challenging questions.

6. Avoiding Negative Wording

Phrase questions and statements positively to minimize confusion.

Avoid using double negatives or negations that can lead to misinterpretation.

7. Avoiding Ambiguity

Ensure that each item has a single, clear interpretation.

Avoid vague or ambiguous language that could lead to multiple interpretations.

8. Appropriate Context

Place items in appropriate contexts and scenarios that are relevant to the subject matter.

Consider real-world scenarios to enhance the authenticity of the assessment.

9. Fairness and Inclusivity

Ensure that items are fair and unbiased, avoiding cultural or gender-specific references.

Use inclusive language to accommodate a diverse group of test-takers.

10. Feedback and Revision

Collect feedback from subject matter experts or colleagues to identify potential issues with items.

Revise items based on feedback and continuously improve the quality of the assessment.

11. Pilot Testing

Before using items in a high-stakes assessment, conduct pilot testing with a small group to identify any flaws or issues with the items.

12. Use of Bloom's Taxonomy:

Consider incorporating a range of cognitive levels from Bloom's Taxonomy (knowledge, comprehension, application, analysis, synthesis, and evaluation) to ensure a comprehensive assessment.

By incorporating these approaches, item writers can develop effective and reliable assessments that accurately measure the intended learning outcomes.

Item Difficulty

The item difficulty is a measure of the proportion of respondent who respond to an item correctly (Nunnally, 1972; Thorndike et al. 1991). The difficulty index is calculated by using following formula: $DV = \text{Total No of correct Response} / N$ The P value of an item difficulty supplied an accurate evidence of how easy or difficult the item was for the responded. The difficulty index can range from 0.00 to +1.00. For a multiple choice test consisting of four or more alternatives, items in the range between 0.20 and 0.80 should be selected (Nunnally, 1972). All items found too easy or too difficult were excluded.

- To examine the difficulty level of test items; whether the difficulty level set up at the time of test construction was right or not.
- To see if there are any items which are correctly attempted by everyone. If there are any such items that are 'too easy' then they need to be removed, replaced, or revised.
- Similarly, if there are any items which are not correctly attempted by anyone then such 'too difficult' items also need to be removed, replaced, or revised.

Item-Difficulty Index

As mentioned earlier, item difficulty index is either in form of percentages or proportions of the total number of test takers who attempted an item correctly. Item difficulty is calculated separately for every item. It is denoted by a lowercase italicized 'p'. A number attached as subscript to this 'p' indicates the item number whose difficulty level is described. For example p₁ indicates item difficulty of item number 1, p₂ is the difficulty level of item number 2 and so on. On occasions you may come across the term 'facility index' rather than difficulty index that refers to the percentage of responses to correct choices. Both terms refer to the same procedure.

Item Discrimination

Discriminatory power of item supplies us information to what extent the test is able to discriminate between high and low achievers on achievement test.

Taking Care of Guessing:

In case of some tests the test taker can give the right answer simply by guessing. This happens mostly in case of items where multiple response options are provided. In such tests the desired or acceptable proportion of accurate answers is set higher than that in free response tests. The optimal average difficulty for MCQ type tests is calculated by taking the mid-point between chance success proportion and 1.00. Chance success proportion is the likelihood of giving a correct response simply by guessing; in MCQs with 4 options it is .25; with 5 options it is .20, and with three options it will be .33. The mid-point can be calculated by adding the chance success proportion and 1.00 and dividing the sum by 2. For example in a test with 4 response options in each item, the chance success proportion is .20. The optimal item difficulty will be: $.20 + 1.00 / 2 = 1.20 / 2 = .6$. For a test with 5 options the average proportion correct should be .69.

Item Difficulty Index In Different Types Of Tests:

Item difficulty index may not be based on the success rate/passing proportion/ number of accurate responses every time. In case of achievement tests or intellectual ability tests the index is based on the proportion or percentage of people who gave the correct answer. However, in case of tests where there are no right or wrong answers, item- endorsement index may be used. Here the analysis can help identify items which nobody replied to, which received the same response from every one etc.

Item Discrimination Index:

A test is supposed to discriminate between those who know and those who do not know; those who score high and those who score low; those who have acquired a skill and those who have not. A test will not be a good test if the people who are supposed to know the correct answer fail and those who are not supposed to know succeed. A good test differentiates between the high and low scorers. If some items are correctly answered by high scorers and some by low scorers then something is wrong with the test.

The discrimination power was calculated using the formula: $DP = RU - RL / N$

Where, RU = Correct response in the upper group

RL = Correct response in the lower group

N = the number of students in either group

For calculating discrimination index researcher arranged response sheets in the descending order of the scores. The top 27% (22 Nos.) response sheets and bottom 27% (22 Nos.) response sheets were used for statistical treatment. The numbers of respondent given correct response in the upper group (U) and in the lower group (L) on same item were identified and subtracted from upper group to lower group who answered the item correctly. The greater the difference, the better the

item as it could discriminate the upper group from the lower group. The items having a DI of below 0.13 are considered as poor discriminators and eliminated from test, however difficulty value is also considered for taking final decision about any item.

Range	Grade	Recommendations
> 0.39	Excellent	Preserve
0.30-0.39	Good	Possibilities for enhancement
0.20-0.29	Average	Need to verify/review
0.00-0.19	Poor	Reject or review in depth (Items having good difficulty value but discrimination index up to 0.10 is considered for revision and finally included in the test)
< -0.01	Worst	Remove

Item Analysis

In this section you will be introduced to the following concepts:

1. Item response theory
2. Item- characteristic curves
3. Cross validation
4. Qualitative analysis of tests

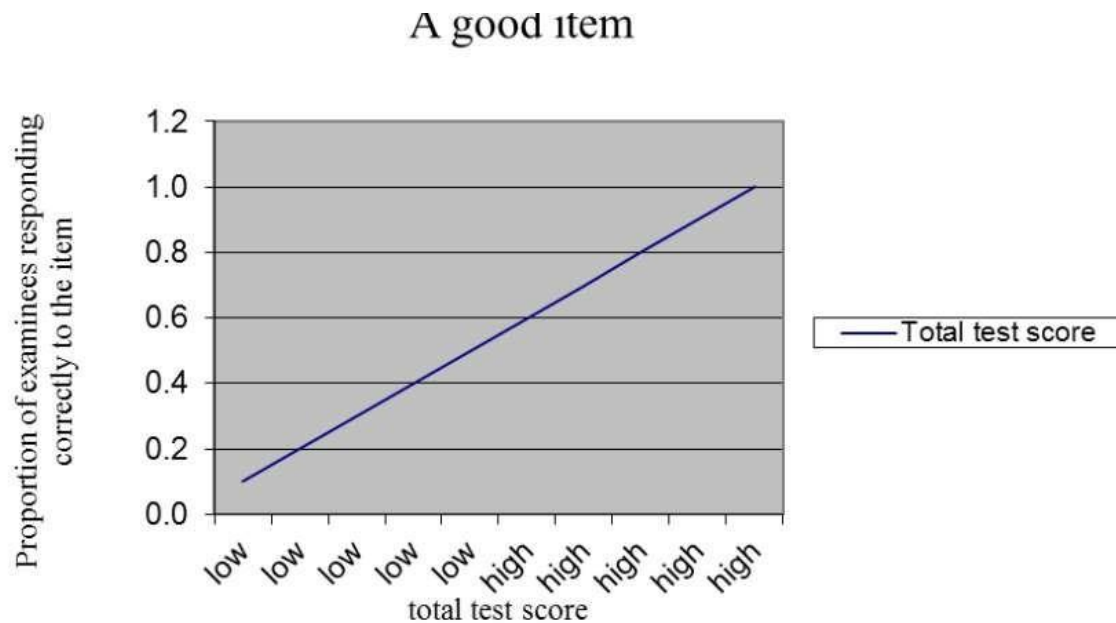
Item Response Theory

Item response theory is an approach that takes into consideration the probability of answering, right or wrong, each individual item in a test. The information regarding each item is plotted graphically. This approach is known as the item response theory or IRT. It is also known as Item-Characteristic Curve Theory and Latent Trait Theory. The graph containing information about the items is called the item- characteristic curve. Decisions regarding the items of a test can be based upon this information.

Item- Characteristic Curves

Item difficulty and item discrimination can be presented graphically also. Item- characteristic curves are the graphs that represent these characteristics of a test. The horizontal axis represents the ability being tested whereas the vertical axis contains the probability correct responses or the proportion of examiners responding correctly to the item. In the words of Kaplan and Saccuzzo (2001), it is “a graph prepared as part of the process of item analysis. One graph is prepared for each item and shows the total test score on the X axis and the proportion of test takers passing the item on Y axis” (p. 637). The shape or slope of the graph or curve indicates whether the item is a

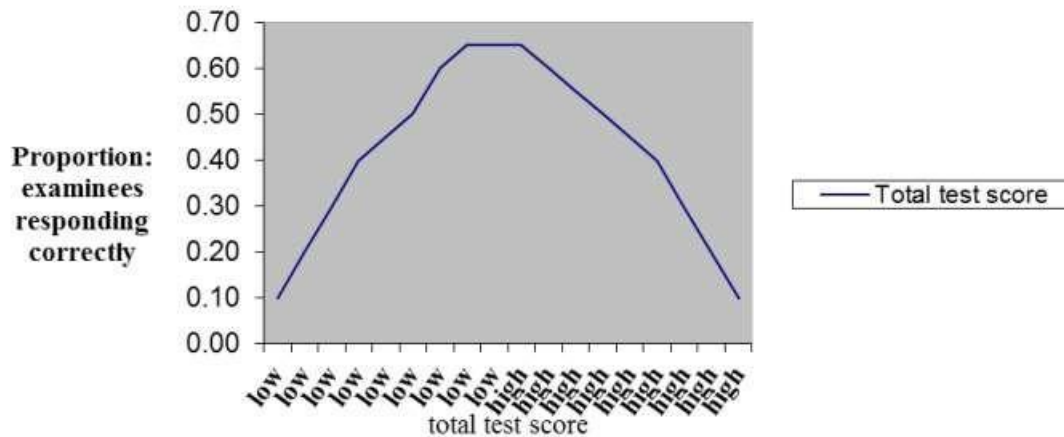
good one or not, how far does it discriminate high scorers from low scorers. A steep slope indicates that the test discriminates between the two groups. Scores of a highly discriminating test will yield a very steep slope. A good item has a positive slope. As can be seen from the following graph, the proportion of high scorers responding correctly is higher than the proportion of low scorers. More of the low scorers are not responding correctly. It can be said that the item that yielded such a curve is a good item



A negative slope like the following one, or a similar one in the same direction, means that the item is not a good one. The item does discriminate between the two groups, high scorers and low scorers, but in the opposite way. We do not expect and accept this type of discrimination. This graph indicates that more of the low scorers did the item correctly than the high scorers. This means that the probability of answering the item correctly is higher for the low scorers rather than the high scorers. This, therefore, is a bad or poor item that needs to be removed or replaced.

Another type of items can be the one in which the majority of neither the top scorers nor the low scorers do the item correctly. It is the middle, moderate, scorers who attain the maximum proportion of correct responses. This type of item is also a bad item

A weak/poor/bad item



Cross validation

Cross validation is a process where a test's validity is confirmed by testing it on different samples than the one used during its creation. This helps ensure that the test accurately predicts outcomes across various groups. If validity is only assessed using the original sample, it might overestimate the test's effectiveness due to chance variations. Through cross validation, we can determine the true validity of the test, which may differ from the initial estimation. Shrinkage, or the decrease in validity when moving from the original sample to a new one, is expected in this process. Factors influencing shrinkage include the size of the original item pool, the proportion of items retained, and the sample size used for validation. A larger item pool and smaller proportion of retained items tend to yield higher validity coefficients, while using larger cross validation samples can reduce shrinkage.

Qualitative Analysis:

Qualitative analysis of a test may also be conducted along with quantitative analysis. After test administration is over, the test takers may be asked questions about various aspects of the test. These questions can be asked and answered orally or in writing. Different formats can be adopted for this purpose e.g. interview, discussions etc. The respondents' responses can be of great help in improving the individual test items, test format, and the entire test itself. Cohen and Swerdlik (1999) have pinpointed some areas that may be explored

- Cultural sensitivity □ Face validity
- Test administrator
- Test environment
- Test fairness
- Test language
- Test length
- Test taker's guessing

- Test taker's integrity
- Test taker's mental/physical state upon entry
- Test taker's mental/physical state during the test
- Test taker's overall impressions
- Test taker's preferences
- Test taker's preparation

